

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.
8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.
9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.
10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 25****Topics covered :**

1. Properties of Fourier Transform
2. Conditions for existence of Fourier Transform

- Let $X(\omega)$ represent the Fourier Transform of a signal $x(t)$, defined as:

$$\mathcal{F}\{x(t)\} = X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt. \quad (1)$$

- **Linearity:** The Fourier Transform preserves linearity, meaning that for any constants a and b ,

$$\mathcal{F}\{ax_1(t) + bx_2(t)\} = aX_1(\omega) + bX_2(\omega). \quad (2)$$

- **Time Shifting:** A shift in the time domain results in a phase shift in the frequency domain:

$$\mathcal{F}\{x(t - t_0)\} = X(\omega)e^{-j\omega t_0}. \quad (3)$$

- **Scaling:** If a function is compressed or stretched in time, its Fourier Transform is inversely stretched or compressed:

$$\mathcal{F}\{x(at)\} = \frac{1}{|a|} X\left(\frac{\omega}{a}\right). \quad (4)$$

- Fourier Transform of $e^{-at}u(t)$: Evaluating the Fourier Transform,

$$\mathcal{F}\{e^{-at}u(t)\} = \int_0^{\infty} e^{-(a+j\omega)t} dt = \frac{1}{a+j\omega}, \quad a \in \mathbb{R}^+. \quad (5)$$

As $a \rightarrow 0^+$, this function approaches a step function in the time domain. However, the corresponding Fourier Transform does not converge in the conventional sense. The formal expression for the Fourier Transform in this case would be:

$$X(\omega) = \int_0^{\infty} e^{-j\omega t} dt. \quad (6)$$

This integral does not converge. Despite this, the limiting case suggests that the Fourier Transform should behave as:

$$\mathcal{F}[u(t)] = \frac{1}{j\omega}. \quad (7)$$

We will investigate this interpretation further later.

- Fourier Transform of $e^{-j\omega_0 t}x(t)$: Evaluating the Fourier Transform,

$$\mathcal{F}\{e^{-j\omega_0 t}x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-j(\omega+\omega_0)t} dt. \quad (8)$$

This results in a frequency shift,

$$\mathcal{F}\{e^{-j\omega_0 t}x(t)\} = X(\omega + \omega_0). \quad (9)$$

In the special case where $x(t) = 1$, we obtain:

$$\mathcal{F}\{e^{-j\omega_0 t}\} = \int_{-\infty}^{\infty} e^{-j(\omega + \omega_0)t} dt. \quad (10)$$

This integral does not converge in the conventional sense, as it does not decay over time. Observing the energy of this signal,

$$E = \int_{-\infty}^{\infty} |e^{-j\omega_0 t}|^2 dt = \int_{-\infty}^{\infty} dt, \quad (11)$$

we see that the signal has infinite energy, confirming its non-convergence. To resolve this, we will use the duality property to obtain its Fourier Transform in a distributional sense. By applying the duality property,

$$\mathcal{F}[e^{-j\omega_0 t}] = 2\pi\delta(\omega + \omega_0). \quad (12)$$

More formally, this result can be derived using the general theory of distribution functions.

- Fourier Transform of $x(t) \cos(\omega_0 t)$: Evaluating the Fourier Transform,

$$\mathcal{F}\{x(t) \cos(\omega_0 t)\} = \int_{-\infty}^{\infty} x(t) \cos(\omega_0 t) e^{-j\omega t} dt. \quad (13)$$

Using Euler's identity,

$$\cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}, \quad (14)$$

we rewrite the Fourier Transform as

$$\mathcal{F}\{x(t) \cos(\omega_0 t)\} = \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]. \quad (15)$$

In the special case where $x(t) = 1$, we obtain:

$$\mathcal{F}\{\cos(\omega_0 t)\} = \frac{1}{2} [\mathcal{F}[e^{j\omega_0 t}] + \mathcal{F}[e^{-j\omega_0 t}]]. \quad (16)$$

Since neither of these individual transforms converge, the function does not have a conventional Fourier Transform. However, using the duality property,

$$\mathcal{F}\{\cos(\omega_0 t)\} = \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]. \quad (17)$$

More formally, this result follows from the general theory of distribution functions.

- Fourier Transform of $e^{at}u(-t)$: Evaluating the Fourier Transform

$$\mathcal{F}\{e^{at}u(-t)\} = \int_{-\infty}^0 e^{(a-j\omega)t} dt = \frac{1}{a - j\omega}, \quad \text{Re}(a) > 0. \quad (18)$$

The Fourier transform of the function $e^{-at}u(t) - e^{at}u(-t)$ is given by

$$\mathcal{F}\{e^{-at}u(t) - e^{at}u(-t)\} = \frac{1}{a + j\omega} - \frac{1}{a - j\omega} = -j \left(\frac{\omega}{a^2 + \omega^2} \right). \quad (19)$$

As $a \rightarrow 0^+$, this function approaches the signum function in the time domain. Thus the Fourier Transform of the Signum Function is given by

$$\mathcal{F}[\text{sgn}(t)] = \frac{2}{j\omega}. \quad (20)$$

- Fourier Transform of the Unit Step Function from signum function: The unit step function can be expressed using the signum function as:

$$u(t) = \frac{1}{2}(1 + \text{sgn}(t)). \quad (21)$$

Taking the Fourier Transform,

$$\mathcal{F}[u(t)] = \frac{1}{2}\mathcal{F}[1] + \frac{1}{2}\mathcal{F}[\text{sgn}(t)]. \quad (22)$$

Using known results,

$$\mathcal{F}[1] = 2\pi\delta(\omega), \quad \mathcal{F}[\text{sgn}(t)] = \frac{2}{j\omega}. \quad (23)$$

Therefore,

$$\mathcal{F}[u(t)] = \pi\delta(\omega) + \frac{1}{j\omega}. \quad (24)$$

This result is different from simply setting $a \rightarrow 0$ in $e^{-at}u(t)$, highlighting the necessity of using distributional techniques in Fourier analysis.

- **Integration:** Let $X(j\omega)$ represent the CTFT of the input signal $x(t)$. The FT of a the signal $y(t) = \int_{-\infty}^t x(t)$ is given by

$$Y(\omega) = \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega) \quad (25)$$